

Vintage Parts India – Complete Project Documentation

1. Problem Statement

In today's automobile market, many vehicle models get discontinued over time. However, a large number of users still own and use these vehicles. The major challenge they face is the **unavailability of spare parts** such as oil filters, air filters, engine components, and body parts.

Currently, there is **no centralized platform** where: - Users can search for discontinued vehicle parts - Retailers or suppliers can list and sell such parts - Buyers and sellers can directly connect

This creates inconvenience, delays, and increased costs for users trying to maintain their vehicles.

2. Proposed Solution

To solve this problem, we propose **Vintage Parts India**, a web-based platform that connects: - **Users (buyers)** looking for discontinued automobile parts - **Retailers/Suppliers (sellers)** who have those parts available

The platform allows: - Sellers to upload and list parts - Buyers to search and connect with sellers - Direct communication between both parties

3. Objectives

- To build a centralized marketplace for discontinued automobile parts
 - To reduce search time and effort for users
 - To help retailers reach potential buyers
 - To create a scalable and user-friendly web platform
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4. Scope of the Project

The system focuses on: - Two-wheeler and four-wheeler vehicle parts - Discontinued or hard-to-find automobile components - Direct buyer-seller interaction (no payment gateway initially)

5. System Users

5.1 Buyer (User)

- Searches for parts
- Views available listings
- Contacts seller

5.2 Seller (Retailer/Supplier)

- Uploads part details
 - Adds price and contact info
 - Manages listings
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6. Features

6.1 Authentication System

- User registration (Sign Up)
- Login functionality

6.2 Landing Page

- Attractive UI
- Introduction to platform

6.3 Category Selection

- Bike
- Four-wheeler

6.4 Product Listing

- View all available parts
- Filter/search functionality (optional enhancement)

6.5 Add Product

Seller can upload: - Image of part - Part name - Vehicle model - Price - Phone number - Address

6.6 Contact System

- Buyer can directly contact seller using provided details
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7. System Architecture

The system follows a **client-server architecture**:

User (Browser) ↓ Frontend (HTML, CSS, JS) ↓ Backend (Node.js / Express) ↓ Database (MongoDB / JSON)

8. Workflow

1. User visits website
 2. Signs up / logs in
 3. Selects category (Bike / Car)
 4. Chooses action:
 5. Search for part
 6. Add a part
 7. If searching:
 8. Browse listings
 9. View details
 10. Contact seller
 11. If adding:
 12. Fill form
 13. Upload image
 14. Submit listing
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9. Technology Stack

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Node.js
- Express.js

Database:

- MongoDB (or JSON storage in basic version)
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10. Data Fields (Product)

- Product Name

- Vehicle Model
 - Category (Bike/Car)
 - Price
 - Image
 - Seller Name
 - Phone Number
 - Address
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11. Advantages

- Solves real-world problem
 - Easy to use interface
 - Direct communication between buyer and seller
 - Scalable into full e-commerce platform
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12. Limitations

- No online payment system
 - No verification of sellers
 - Limited security in basic version
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13. Future Enhancements

- Payment gateway integration
 - Chat system
 - AI-based search recommendations
 - Admin panel
 - Rating & review system
 - Location-based filtering
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14. Conclusion

Vintage Parts India provides an effective solution to the problem of unavailable discontinued automobile parts. It bridges the gap between buyers and sellers and has strong potential to evolve into a full-scale marketplace.

15. Team Members

- Nishit Bhalerao

- (Add others if needed)

16. Use Case Diagram (Textual)

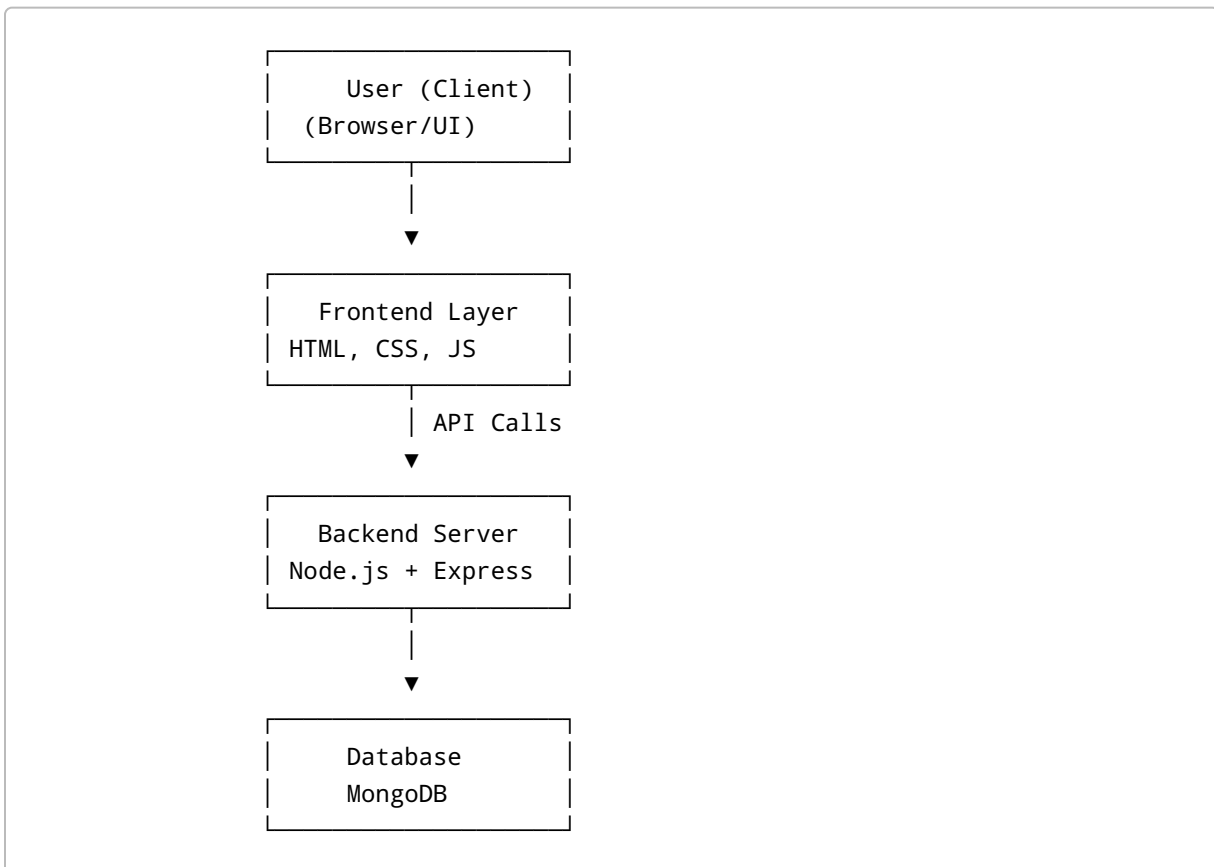
Actors: - User (Buyer) - Seller

Use Cases: - Register/Login - Search Parts - View Parts - Add Parts - Contact Seller

17. Future Scope

This project can be expanded into: - Mobile application - Nationwide marketplace - Integration with automobile service centers

18. System Architecture Diagram (Detailed)

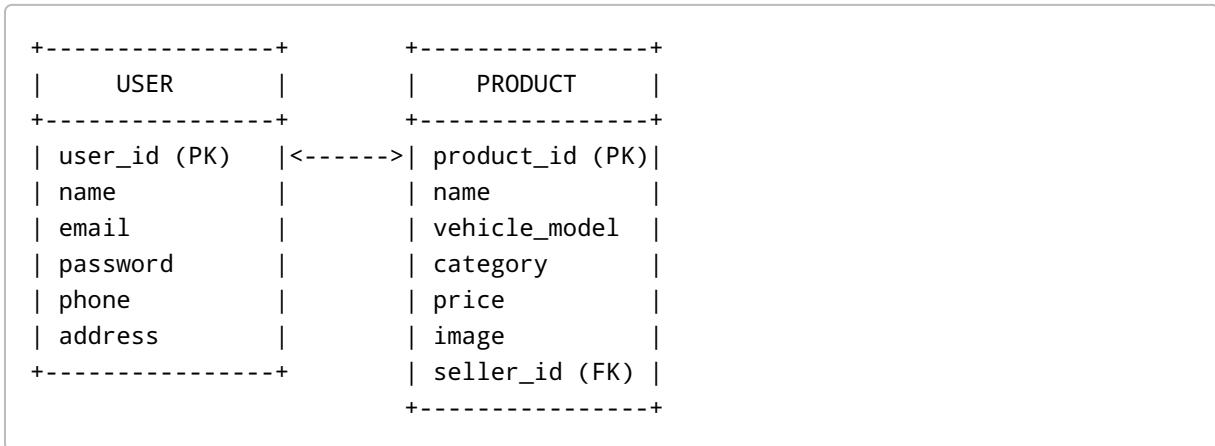


Explanation:

- The **frontend** handles UI and sends requests

- The **backend** processes logic and handles APIs
- The **database** stores user and product data

19. ER Diagram (Entity Relationship Diagram)



Relationships:

- One **User (Seller)** can upload multiple **Products**
- Each **Product** belongs to one **User**

20. Database Design Explanation

User Collection:

- Stores login and personal details
- Used for authentication

Product Collection:

- Stores all part listings
- Linked with user using seller_id

End of Documentation